

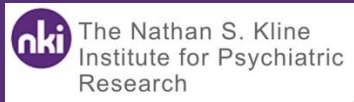


Office of
Mental Health

9.4T MRI scanner: update on the development at NKI

Alexandre R. Franco, PhD

May 22, 2026



The Specifications

The Static Magnet

The "Powerhouse" of the system. It polarizes hydrogen nuclei to create the raw signal. Higher Tesla field strengths provide the SNR necessary for micro-structural imaging.

9.4
Tesla (B0)
Signal Polarization

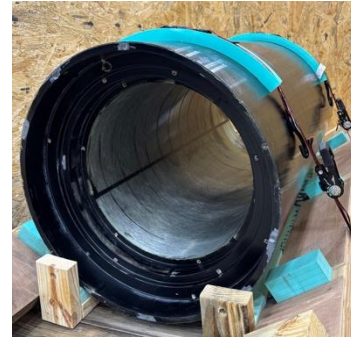


The Gradient System

The "Architect" of the scan. It uses spatial variations in the magnetic field to encode position. Its performance defines the sharpness, speed, and fidelity of the image.

140
mT/m (Gmax)
Resolution Amplitude

810
T/m/s (SR)
Encoding Velocity



Analogy: The Space Observatory



9.4T: The Mirror

The "Light Bucket." A massive primary mirror captures more signal from the darkness of the brain, increasing raw visibility.



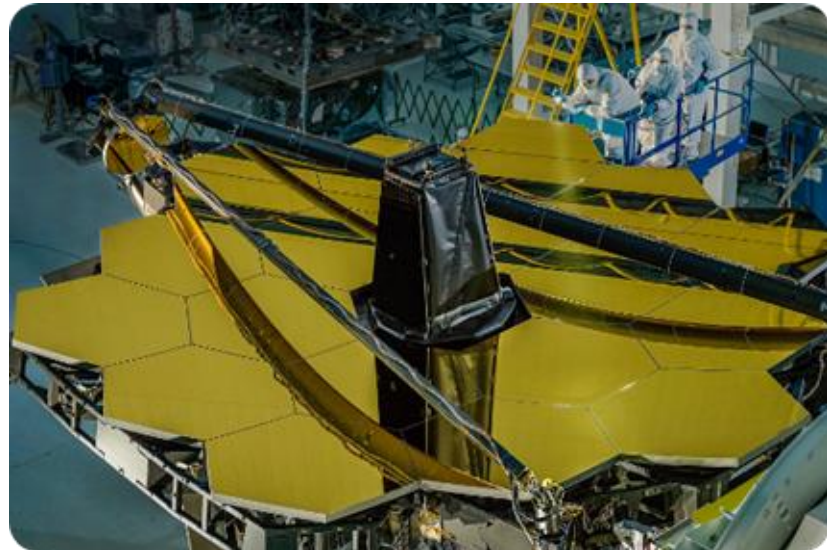
140 mT/m: Zoom

The "Resolving Power." High gradient strength allows the telescope to **magnify and focus** on sub-millimeter stellar structures.



810 SR: Guidance

The "Tracking System." The ultra-fast slew rate ensures the guidance system can lock onto targets without blur during movement.

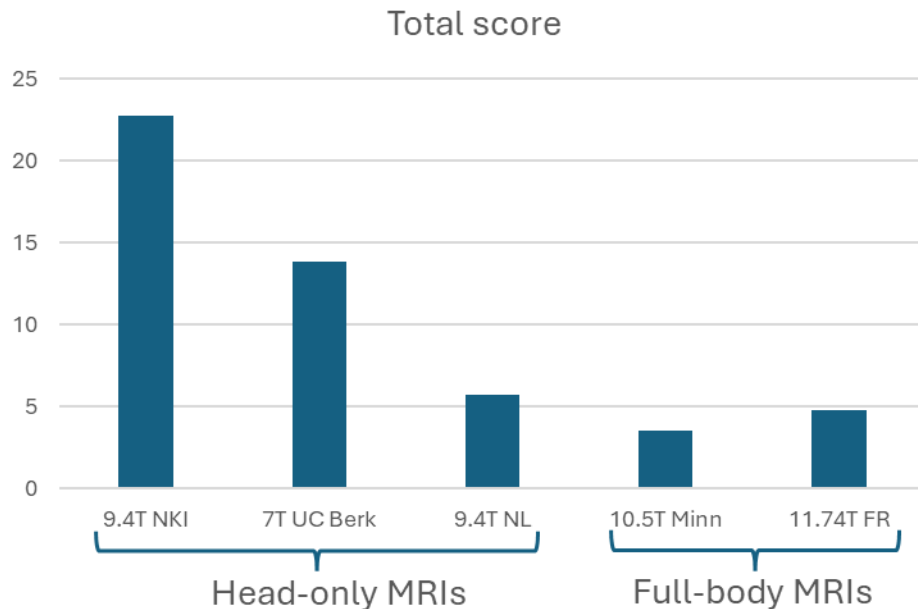


Comparative Metrics

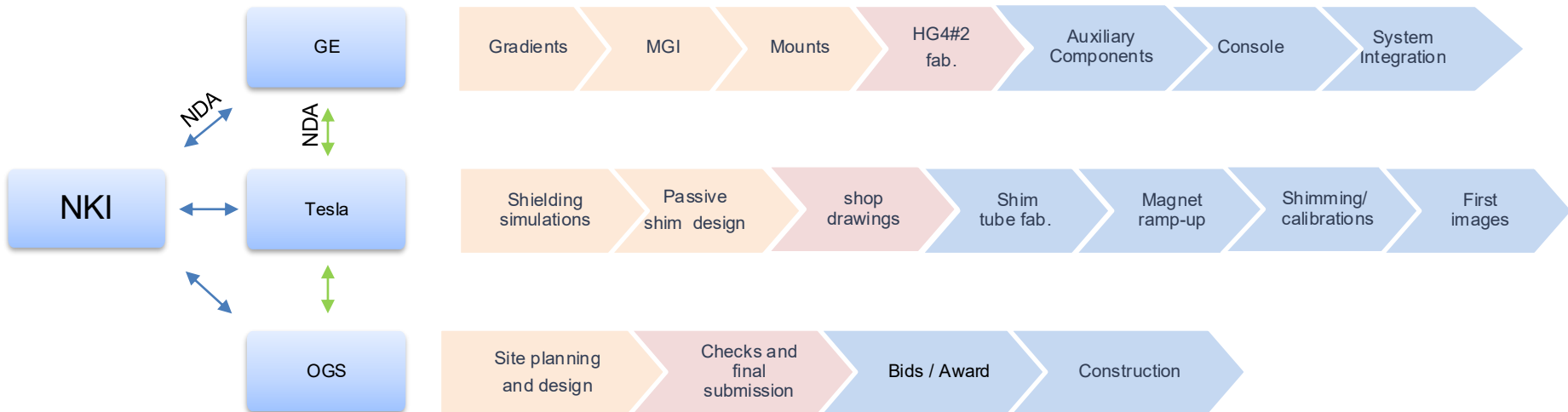
Capability	Standard Clinical (3T)	Neuro-UHF (9.4T)
Field Strength	3.0 Tesla	9.4 Tesla
Peak Gradient Amplitude	45 - 80 mT/m	140 mT/m
Max Slew Rate	200 T/m/s	810 T/m/s
Resolution Limit	~1.0 mm (Isotropic)	< 0.2 mm (Isotropic)
Functional MRI	2.4 mm (iso) 800ms (TR)	<1.0 mm (iso) 100ms (TR)

How does the proposed NKI 9.4T compare to other UHF MRIs?

Scanner name	9.4T NKI	10.5T Minn	11.74T FR	7T UC Berk	9.4T NL
Field strength [T]	9.4	10.5	11.74	7	9.4
uiSNR	1.9	2.4	2.9	1.1	1.9
Gradients strength [mT/m]	150	70	70	200	80
Slew rate [T/m/s]	850	200	200	900	400
Total score	22.7715	3.528	4.76644	13.86	5.7152



Overview of Development



1. Development of RF Coils
2. Development of Data Management Network
3. Communications Power Corp.
4. RF and Safety Lab

9.4T Site



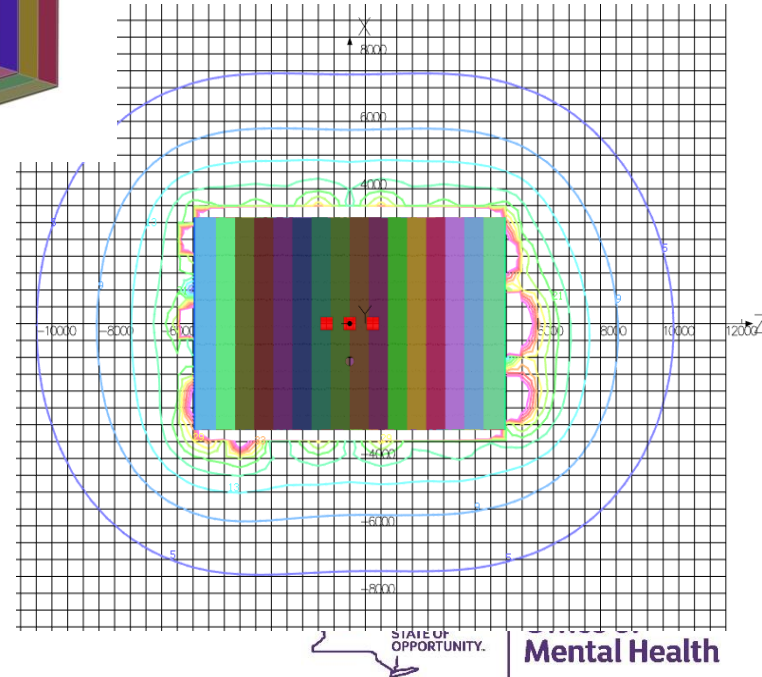
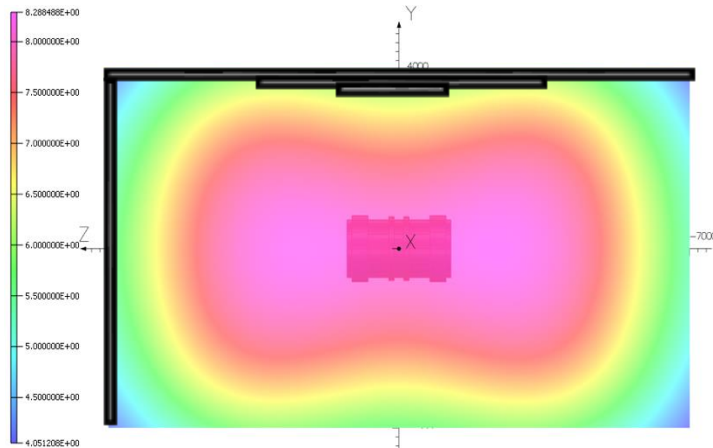
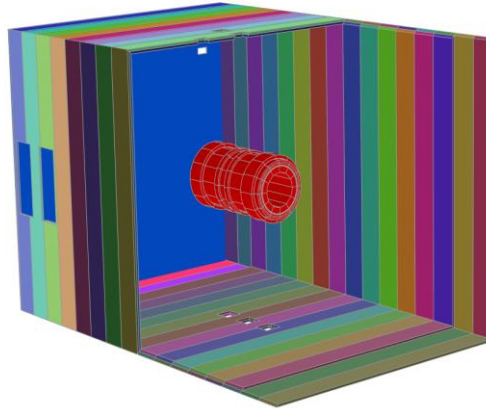
Current schedule:

- 100% submission for review and comment: *August 2025*
- Final Submission Check: *June 2026*
- Bidding for Construction: *Aug 2026-Dec 2025*
- Award for Contractor: *September 2025*
- Start construction: *September 2026*
- Substantial completion: *January 2028*
- Finalize construction: *April 2028*



Fringe field Simulations

- Tesla performed simulations for Office of General Services
- 5 G line c.a. 10 m axially and c.a. 7.5 m radially
- 9 G line inside the equipment room and control room
- Shield weight reduced from 500 tons to 350 tons now



Cryogenics

- Cryocoolers and cryo-compressors
- Pre-cooling system
- Reliquefaction system for zero boiloff

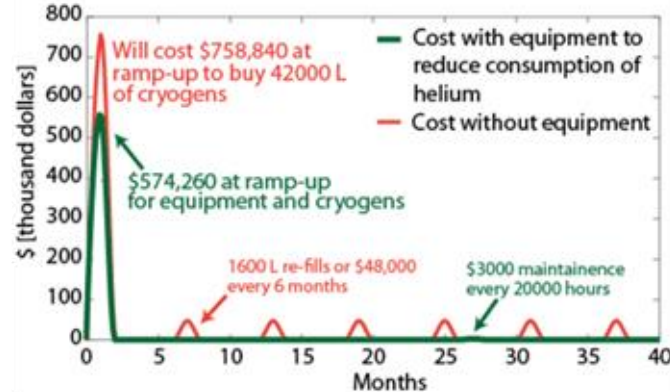


Fig. 6: Photograph of the helium reliquification system

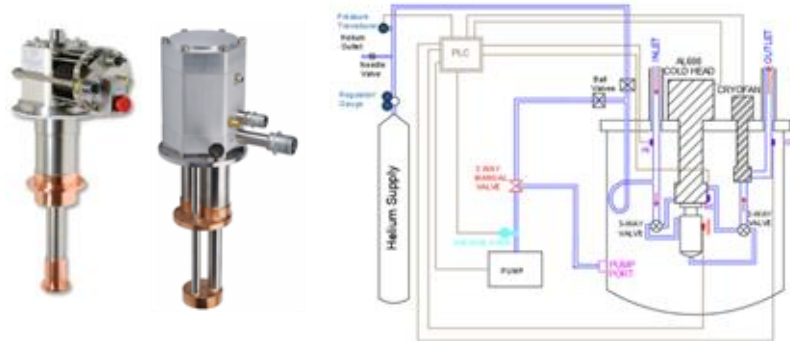


Fig. 4: Working schematic of MPC600 pre-cooler

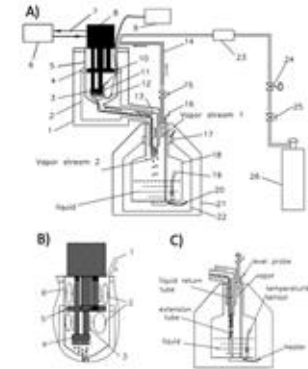
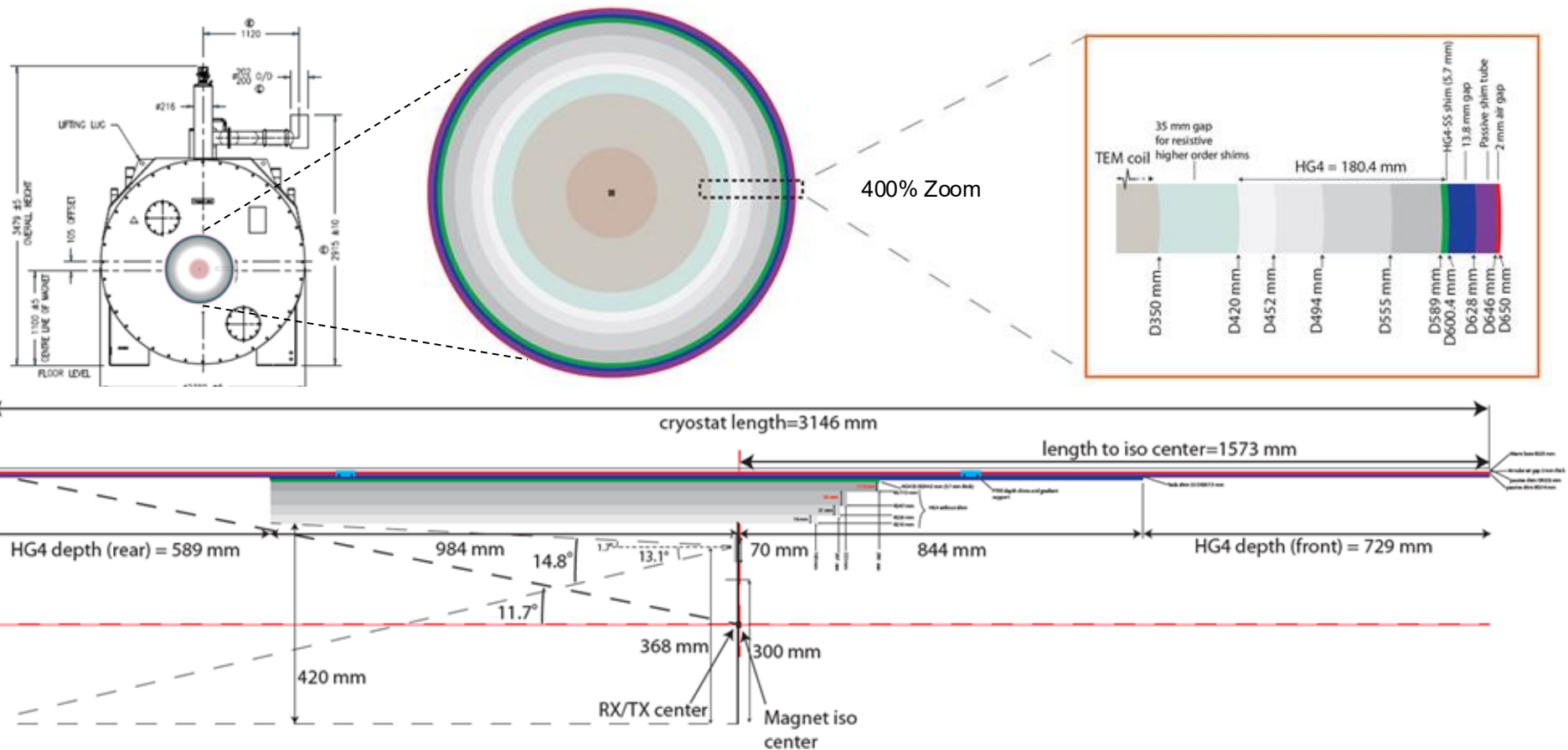


Fig. 7: A) Reliquifier system schematic. B) Helium reliquification in the neck. C) Extension tube of the reliquifier.

Shim tube



Electronics

GE Healthcare



GE Healthcare

Components from their SIGNA 7.0T

Provide a set of SIGNA components including:

- Advanced Gradient Coil and Driver
 - 120mT/m
 - 900T/m/s
 - Optimized Shielding
 - Enhanced Cooling System
 - 900 A/2200 V operating power
- Multichannel Receive Chain
 - Digital Signal processing
 - 64 Simultaneous receive channels
 - Support for at least 32 channel Multinuclear arrays
- Multichannel RF Transmit Chain
 - Two or Eight Channel Multimodal
 - Independent Waveform Synthesis
 - Small signal RF and control interface
- Advanced scanning capabilities including:
 - Compressed Sensing
 - Hyperband simultaneous multislice imaging



**Office of
Mental Health**

RF Amplifiers – Communications Power Corporation

PRELIMINARY

MULTI CHANNEL – B1 SHIMMING

NMR/MRI PLUS PULSED RF POWER AMPLIFIER SYSTEMS MODEL 9T2000M-8C

CPC

80 Davids Drive, Suite 3, Housatonic, New York 11788
Phone: 631.687.7356 ext 224 Fax: 631.634.7025
email: info@CPCAMPS.COM
Web Site: www.CPCAMPS.COM

SPECIFICATIONS - 8 CHANNEL**

Operating frequency	30-410 MHz
Pulse power into 50Ω	2000W (min.) / channel
Gain	66dB nominal
Gain Flatness:	±3dB (rated Pin-15dB)
Gain Linearity	±1.5 dB
Phase Linearity	±1° max excursion
Phase Error over Pulse:	±5° over 10 ms rectangular pulse
Input/Output Impedance:	50Ω nominal (by design)
Input VSWR:	≤2:1
Noise Figure:	12dB typical
Output Noise (blanked):	≤20 dB over thermal
Blanking Delay:	≤2.0 μs
RF Pulse Width:	20msec
Duty Cycle:	≤10%

SYSTEM FEATURES

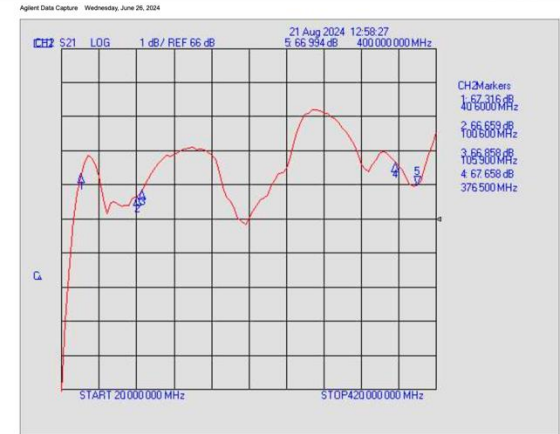
Protection functions: Maximum forward peak power Maximum average power Maximum output VSWR Internal Over-temperature Internal power supply monitoring Controls, front panel: AC power on/off (Circuit Breaker) Forward/ Reflected RF Power Fault reset Front panel indicators: AC power On Standby mode Blanking (Gate) RF overdrive fault Power supply fault status VSWR fault Thermal fault Energy fault Interface functions: (C= Control Input, F= Flag Output) RF power (F flag line) Standby (C/F) RF overdrive (F) VSWR (F) CBI mode (C/F) Power supply (F) Thermal (F) Energy (F)	User Connections, rear panel: AC mains, Receptacles, EMI filtered PA RF input (x8): SMA (F) PA RF output (x8): Type N (F) (x2 type NFF) Noise blanking 25 pin Dsub System interface RS232 Protective ground: threaded bolts ENVIRONMENTAL Cooling: Internal forced air, front to rear Operating temperature: +10 to 40 °C AC line voltage: 200-240 VAC, 1 phase, ±10%, 47-63 Hz Chassis size: 71" H X 23" W X 33" D HW: inches Net weight: 1200 lbs. (approx.) Compliance: CE, UL 2601 upon request
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8/78/11



Multi-channel, multi-frequency, multi-stage power amplifier system

July, 2024



Office of
Mental Health

Stakeholders - Major



NKI

Vaughan
Franco
Zia
Roe
Milham
Colcombe
Goff
Schroeder
Russ

COLUMBIA
UNIVERSITY



Columbia

Vaughan
Ray Lee
Sairam Geethanath
Christoph Juchem



NYU Langone

Dan Sodickson
Guillaume Madelin
Ivan Kirov
Ryan Brown
Zhe (Sage) Chen



GE Healthcare

Thomas Foo (Gradients)
Rob Peters (7T)



Mt. Sinai

Haroutunian
Roussos

Yale

Yale

Robin de Graaf



CUNY

Lucas Parra

50+ Researchers in the tristate area expressed their commitment and intent to use the 9.4T



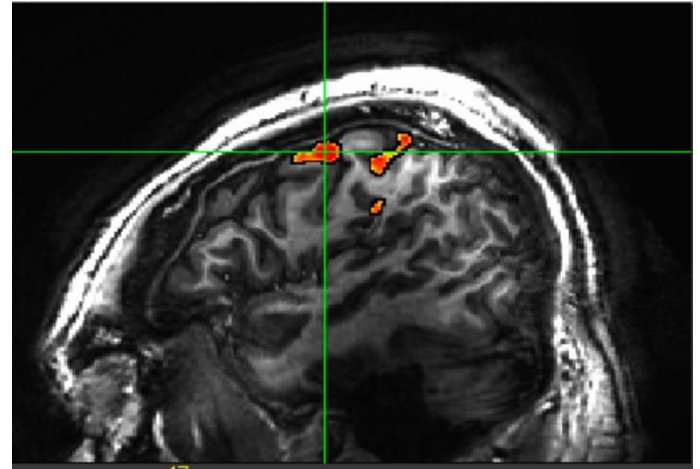
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Mental Health

Progress of Development of 9.4T MRI at NKI

Task	Description	Comments	2022				2023				2024				2025				2026				2027				2028				2029			
			Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec	Jan-Mar	Apr-June	Jul-Sept	Oct-Dec				
Building design and construction	Design																																	
	Final submission check																																	
	Bids for Construction																																	
	Awarding Contractor																																	
	Construction																																	
Magnet-Gradient Interaction	Obtain data on 9.4T magnet with details of positions of magnet coil elements, thermal shield, and dimensions																																	
	Perform simulations of MGI to determine if gradient coil switching will generate acceptable heat load on magnet.	Information is important to ensure that magnet does not quench, and assess eddy current field in the imaging volume																																
	Optimized MGI	perform MGI again																																
Passive shim design	Determine passive shim requirement after superconducting shim. This affects shim tray design and gradient coil mounting.	(1) Assign resources to perform/optimize superconducting shim (2) Estimate residual B0 (3) Run simulation for passive shim requirement (Max force on shim trays, radial space, number of shim trays)																																
	Passive shim optimization																																	
Passive shim fabrication																																		
Gradient/Magnet integration	Design: (1)Gradient mounts (2)Bed bridge and rails (3)Coolant manifold (4)Gradient bus connectors (5)Wire and hose guides																																	
	Gradient/magnet vibration																																	
	Fabrication of parts required for mounting																																	
Magnet/gradient installation	Installation of magnet at NKI	Completion of building and relocation of magnet to NKI campus																																
	Installation of gradient coil	Installation of gradient coil and securing of mounting before magnet is ramped																																
	Magnet cooldown, ramp, and shim	Will be done by Tesla Engineering They will need a field camera that will fit in the gradient bore of 42-cm ID																																
Systems integration	Delivery of systems electronics	GE SIGNA 7T electronics component																																
	Delivery of RF amplifier specific to 9.4T system	Provided by Communication Power Corporation (CPC)																																
	Frequency Conversion	Development and fabrication of electronics to convert signal from 9.4T to 3T																																
	Systems integration and testing	Integration of all electronics components Ensure compatibility with latest software																																
	RF transmit/receive coil - design, fabrication, FDA IDE, and testing																																	
	First phantom image																																	
	Systems configuration and calibrations	Checking systems configuration and performance - eddy current calibrations, concomitant gradient field calibrations, gradient linearity calibrations																																
Safety assessments	Testing for PNS	Can be done before magnet ramp																																
	Testing for acoustics																																	
	SAR simulations and validation	SAR modeling will need to be performed and validated.																																
Imaging studies	Pulse sequence optimizations	Making sure that there are no errors with the pulse sequences with slew rate or gradient amplitude																																
	First human images																																	

completed
active
future tasks

Thank you,
Questions?



Alexandre R. Franco, PhD

May 22, 2026